Request Packet

```
Frame 2744: Packet, 518 bytes on wire (4144 bits), 518 bytes captured (4144 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-A952
    Section number: 1
    Interface id: 0 (\Device\NPF_{C68BC7DB-BFEE-42F5-A952-3C3CBACB27DF})
        Interface name: \Device\NPF_{C68BC7DB-BFEE-42F5-A952-3C3CBACB27DF}
        Interface description: Ethernet 2
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov  6, 2025 07:18:35.370357000 Pacific Standard Time
    UTC Arrival Time: Nov  6, 2025 15:18:35.370357000 UTC
    Epoch Arrival Time: 1762442315.370357000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 2.353000 milliseconds]
    [Time delta from previous displayed frame: 4.155000 milliseconds]
    [Time since reference or first frame: 9 minutes, 18.792221000 seconds]
    Frame Number: 2744
    Frame Length: 518 bytes (4144 bits)
    Capture Length: 518 bytes (4144 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
Ethernet II, Src: HP_4a:84:68 (24:fb:e3:4a:84:68), Dst: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
Internet Protocol Version 4, Src: 172.18.175.248, Dst: 146.190.62.39
Transmission Control Protocol, Src Port: 51777, Dst Port: 80, Seq: 1, Ack: 1, Len: 464
Hypertext Transfer Protocol
```

```
Frame 2744: Packet, 518 bytes on wire (4144 bits), 518 bytes captured (4144 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-A952-3
Ethernet II, Src: HP_4a:84:68 (24:fb:e3:4a:84:68), Dst: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
    Destination: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
    Source: HP_4a:84:68 (24:fb:e3:4a:84:68)
    Type: IPv4 (0x0800)
    [Stream index: 1]
Internet Protocol Version 4, Src: 172.18.175.248, Dst: 146.190.62.39
Transmission Control Protocol, Src Port: 51777, Dst Port: 80, Seq: 1, Ack: 1, Len: 464
Hypertext Transfer Protocol
```

```
Frame 2744: Packet, 518 bytes on wire (4144 bits), 518 bytes captured (4144 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-A952-3
Ethernet II, Src: HP_4a:84:68 (24:fb:e3:4a:84:68), Dst: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
Internet Protocol Version 4, Src: 172.18.175.248, Dst: 146.190.62.39
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 504
    Identification: 0x328b (12939)
    010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.18.175.248
    Destination Address: 146.190.62.39
    [Stream index: 55]
Transmission Control Protocol, Src Port: 51777, Dst Port: 80, Seq: 1, Ack: 1, Len: 464
Hypertext Transfer Protocol
```

```
Transmission Control Protocol, Src Port: 51777, Dst Port: 80, Seq: 1, Ack: 1, Len: 464
    Source Port: 51777
    Destination Port: 80
    [Stream index: 28]
    [Stream Packet Number: 4]
  > [Conversation completeness: Complete, WITH_DATA (63)]
    [TCP Segment Len: 464]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 840483863
    [Next Sequence Number: 465    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 3645162674
    0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
    Window: 1026
    [Calculated window size: 262656]
    [Window size scaling factor: 256]
    Checksum: 0x2edb [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (464 bytes)
> Hypertext Transfer Protocol
```

```
> Frame 2744: Packet, 518 bytes on wire (4144 bits), 518 bytes captured (4144 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-A952-3
> Ethernet II, Src: HP_4a:84:68 (24:fb:e3:4a:84:68), Dst: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
> Internet Protocol Version 4, Src: 172.18.175.248, Dst: 146.190.62.39
> Transmission Control Protocol, Src Port: 51777, Dst Port: 80, Seq: 1, Ack: 1, Len: 464
v Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
    Host: httpforever.com\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchar
    Referer: https://www.google.com/\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Response in frame: 2749]
    [Full request URI: http://httpforever.com/]
```

Response Packet

```
Frame 3138: Packet, 1331 bytes on wire (10648 bits), 1331 bytes captured (10648 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-
    Section number: 1
    Interface id: 0 (\Device\NPF_{C68BC7DB-BFEE-42F5-A952-3C3CBACB27DF})
        Interface name: \Device\NPF_{C68BC7DB-BFEE-42F5-A952-3C3CBACB27DF}
        Interface description: Ethernet 2
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov  6, 2025 07:18:36.729160000 Pacific Standard Time
    UTC Arrival Time: Nov  6, 2025 15:18:36.729160000 UTC
    Epoch Arrival Time: 1762442316.729160000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 1.057000 milliseconds]
    [Time delta from previous displayed frame: 8.396000 milliseconds]
    [Time since reference or first frame: 9 minutes, 20.151024000 seconds]
    Frame Number: 3138
    Frame Length: 1331 bytes (10648 bits)
    Capture Length: 1331 bytes (10648 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:xml]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
Ethernet II, Src: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b), Dst: HP_4a:84:68 (24:fb:e3:4a:84:68)
Internet Protocol Version 4, Src: 146.190.62.39, Dst: 172.18.175.248
Transmission Control Protocol, Src Port: 80, Dst Port: 51778, Seq: 1, Ack: 417, Len: 1277
Hypertext Transfer Protocol
```

```
Frame 3138: Packet, 1331 bytes on wire (10648 bits), 1331 bytes captured (10648 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-A9
Ethernet II, Src: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b), Dst: HP_4a:84:68 (24:fb:e3:4a:84:68)
    Destination: HP_4a:84:68 (24:fb:e3:4a:84:68)
    Source: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
    Type: IPv4 (0x0800)
    [Stream index: 1]
Internet Protocol Version 4, Src: 146.190.62.39, Dst: 172.18.175.248
Transmission Control Protocol, Src Port: 80, Dst Port: 51778, Seq: 1, Ack: 417, Len: 1277
Hypertext Transfer Protocol
eXtensible Markup Language
```

```
Frame 3138: Packet, 1331 bytes on wire (10648 bits), 1331 bytes captured (10648 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-A9
Ethernet II, Src: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b), Dst: HP_4a:84:68 (24:fb:e3:4a:84:68)
Internet Protocol Version 4, Src: 146.190.62.39, Dst: 172.18.175.248
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1317
    Identification: 0x2e55 (11861)
    010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 47
    Protocol: TCP (6)
    Header Checksum: 0xeb8d [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 146.190.62.39
    Destination Address: 172.18.175.248
    [Stream index: 55]
Transmission Control Protocol, Src Port: 80, Dst Port: 51778, Seq: 1, Ack: 417, Len: 1277
Hypertext Transfer Protocol
eXtensible Markup Language
```

```
Transmission Control Protocol, Src Port: 80, Dst Port: 51778, Seq: 1, Ack: 417, Len: 1277
    Source Port: 80
    Destination Port: 51778
    [Stream index: 29]
    [Stream Packet Number: 6]
  > [Conversation completeness: Complete, WITH_DATA (63)]
    [TCP Segment Len: 1277]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 925496507
    [Next Sequence Number: 1278    (relative sequence number)]
    Acknowledgment Number: 417    (relative ack number)
    Acknowledgment number (raw): 3296614134
    0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
    Window: 501
    [Calculated window size: 64128]
    [Window size scaling factor: 128]
    Checksum: 0x6bec [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (1277 bytes)
> Hypertext Transfer Protocol
```

```
> Frame 3138: Packet, 1331 bytes on wire (10648 bits), 1331 bytes captured (10648 bits) on interface \Device\NPF_{C68BC7DB-BFEE-42F5-A9
> Ethernet II, Src: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b), Dst: HP_4a:84:68 (24:fb:e3:4a:84:68)
> Internet Protocol Version 4, Src: 146.190.62.39, Dst: 172.18.175.248
> Transmission Control Protocol, Src Port: 80, Dst Port: 51778, Seq: 1, Ack: 417, Len: 1277
v Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Server: nginx/1.18.0 (Ubuntu)\r\n
    Date: Thu, 06 Nov 2025 07:18:36 GMT\r\n
    Content-Type: image/svg+xml\r\n
  > Content-Length: 451\r\n
    Last-Modified: Wed, 16 Nov 2022 20:59:46 GMT\r\n
    Connection: keep-alive\r\n
    ETag: "63754f42-1c3"\r\n
    Referrer-Policy: strict-origin-when-cross-origin\r\n
    X-Content-Type-Options: nosniff\r\n
    Feature-Policy: accelerometer 'none'; camera 'none'; geolocation 'none'; gyroscope 'none'; magnetometer 'none'; microphone 'none';
    […] Content-Security-Policy: default-src 'self'; script-src cdnjs.cloudflare.com 'self' 'report-sha256'; style-src cdnjs.cloudflar
    Accept-Ranges: bytes\r\n
    \r\n
    [Request in frame: 2945]
    [Time since request: 228.050000 milliseconds]
    [Request URI: /css/images/header-major-on-light.svg]
    [Full request URI: http://httpforever.com/css/images/header-major-on-light.svg]
    File Data: 451 bytes
```

The first protocol is the physical layer where how many bits are going at a time, from which interface they are coming or going are shown, also the frame size is also shown.

The second protocol is the data link layer which has the mac address(a 12 digit hexadecimal number like Destination MAC: c0:f6:ec:90:a9:0b (server's interface MAC).
- Source MAC: 24:fb:e3:4a:84:68 (client's NIC MAC). Also the type IPv4 is shown

The third protocol is the Network layer where it provides logical addressing like - Source IP: 172.18.175.248 (client address). - Destination IP: 146.190.62.39 (server address). It also uses checksum for error detection.TTL is basically time to live, how long a packet is to be alive or a lifetime of it.

The fourth protocol is the transport layer protocol (TCP) which has the segmentation sequence , acknowledgement number which defines from which number the previous segmentation of the received packet was done which is 1 in this case. The port numbers to define the processes, - Source Port: 51777 (randomly chosen by client). - Destination Port: 80 (HTTP standard port). The flags "PSH,ACK" indicate that a TCP packet is simultaneously acknowledging receipt of previous data and pushing the current data to the receiving application.

The fifth protocol is the application layer protocol HTTP. the version is 1.1, 200 from the receiver side means the sending the data was successful, using the GET method on the sender side, Ubuntu is the user agent which indicates that ubuntu was used to send the request, host is the domain name which is keepforever.content length means the size of the sent data's body in bytes which is 451. E tag is basically a specific tag that is used to determine the sent file uniquely. It changes if the modified date is crossed.